# WeatherAdjusted Solar Weather-Adjusted Performance Audit

**Site:** Sample site: 5 MW AC ground-mount, northern Indiana **Period:** 2024-01 to 2025-12 **Report date:** Nov 11, 2026

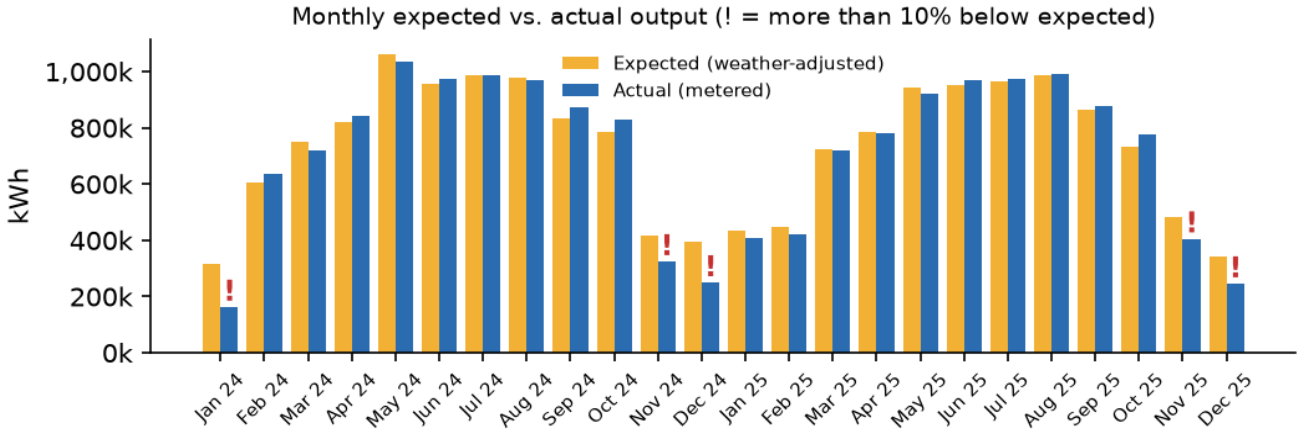
## Summary

In the 12 snow-free months, after adjusting for the actual weather, this site produced **126,294 kWh (1.1%) more** than expected. 0 of those months were more than 10% below expected, so **no under-production was found**.

**Snow months:** 12 months had possible snow (NASA POWER), with a combined gap of 606,543 kWh and 5 flagged. Snow isn't in the model, so these months are shown but left out of the headline loss.

All expected values and dollar figures are **estimates** from a public model and satellite weather data. See Methods.

| | | | |
|---|---|---|---|
| System | 6,400 kW DC / 5,000 kW AC | Mounting | Fixed (open rack) |
| Location | Northern Indiana (exact location withheld) | Tilt / azimuth | 25° / 180° |
| Actual data | EIA-923 Final 2024 and 2025 (public monthly survey data), net generation. kWh = published MWh x 1000. | Energy price | $0.139/kWh (Estimate: EIA Indiana average commercial rate, Aug 2025. Not this site's actual rate.) |



## Monthly results

| Month | Typical-year kWh | Weather ratio | Expected kWh | Actual kWh | Gap kWh | Gap % | Est. lost $ | Snow? | Flag |
|---|---|---|---|---|---|---|---|---|---|
| 2024-01 | 356,924 | 0.879 | 313,806 | 161,000 | 152,806 | 48.7% | $21,179 | yes | YES |
| 2024-02 | 475,416 | 1.277 | 607,191 | 636,000 | -28,809 | -4.7% | -$3,993 | yes | |
| 2024-03 | 720,129 | 1.041 | 749,749 | 718,000 | 31,749 | 4.2% | $4,400 | yes | |
| 2024-04 | 806,726 | 1.015 | 818,675 | 844,000 | -25,325 | -3.1% | -$3,510 | yes | |
| 2024-05 | 941,840 | 1.126 | 1,060,219 | 1,034,000 | 26,219 | 2.5% | $3,634 | | |
| 2024-06 | 933,167 | 1.025 | 956,494 | 975,000 | -18,506 | -1.9% | -$2,565 | | |
| 2024-07 | 1,025,365 | 0.963 | 987,842 | 986,000 | 1,842 | 0.2% | $255 | | |
| 2024-08 | 928,803 | 1.056 | 980,489 | 970,000 | 10,489 | 1.1% | $1,454 | | |
| 2024-09 | 777,996 | 1.069 | 831,961 | 872,000 | -40,039 | -4.8% | -$5,549 | | |
| 2024-10 | 616,677 | 1.272 | 784,410 | 828,000 | -43,590 | -5.6% | -$6,042 | | |
| 2024-11 | 464,694 | 0.9 | 418,220 | 323,000 | 95,220 | 22.8% | $13,197 | yes | YES |
| 2024-12 | 411,433 | 0.958 | 394,219 | 248,000 | 146,219 | 37.1% | $20,266 | yes | YES |
| 2025-01 | 356,924 | 1.214 | 433,466 | 409,000 | 24,466 | 5.6% | $3,391 | yes | |
| 2025-02 | 475,416 | 0.937 | 445,595 | 420,000 | 25,595 | 5.7% | $3,547 | yes | |
| 2025-03 | 720,129 | 1.005 | 723,824 | 718,000 | 5,824 | 0.8% | $807 | yes | |
| 2025-04 | 806,726 | 0.972 | 783,908 | 782,000 | 1,908 | 0.2% | $264 | yes | |
| 2025-05 | 941,840 | 1.003 | 944,820 | 921,000 | 23,820 | 2.5% | $3,301 | | |
| 2025-06 | 933,167 | 1.018 | 950,187 | 968,000 | -17,813 | -1.9% | -$2,469 | | |
| 2025-07 | 1,025,365 | 0.943 | 966,722 | 975,000 | -8,278 | -0.9% | -$1,147 | | |
| 2025-08 | 928,803 | 1.064 | 988,431 | 993,000 | -4,569 | -0.5% | -$633 | | |
| 2025-09 | 777,996 | 1.113 | 866,178 | 876,000 | -9,822 | -1.1% | -$1,361 | | |
| 2025-10 | 616,677 | 1.187 | 731,953 | 778,000 | -46,047 | -6.3% | -$6,382 | | |
| 2025-11 | 464,694 | 1.034 | 480,520 | 404,000 | 76,520 | 15.9% | $10,606 | yes | YES |
| 2025-12 | 411,433 | 0.837 | 344,370 | 244,000 | 100,370 | 29.1% | $13,911 | yes | YES |
| **Total** | | | **17,563,249** | **17,083,000** | **480,249** | **2.7%** | **$66,562** | | **5** |

Negative gaps mean the site beat the weather-adjusted expectation. Weather ratio = actual monthly sunlight ÷ long-term average for that month (NASA POWER). Snow? = NASA POWER snow depth over 1 cm or snowfall over 0.1 mm/day that month.

## Possible causes to check

• 5 of 5 flagged months had possible snow per NASA POWER. **Snow cover** is the most likely explanation for those and isn't in the model, so check local snow records before suspecting equipment.

• No snow-free month was flagged, so this data gives no sign of an equipment problem.

• Monthly data shows *that* a site under-produced and *when*, not *why*. Confirming a cause needs inverter-level data or a site check by the owner's O&M provider.

## Next steps

1. Compare the flagged winter months with local snowfall and snow-cover records.  2. If those months had little snow, pull inverter-level data for them.  3. Otherwise no equipment follow-up is suggested from this data.

# Methods: How WeatherAdjusted Solar Estimates Weather-Adjusted Under-Production

**What this report answers:** Did this site produce less than it should have, given the weather it actually had? If so, roughly how much energy and money was lost?

### *1. Baseline: what a healthy system makes in a typical year*

We model the site in **PVWatts v8**, NREL's public solar performance model (now hosted by the National Laboratory of the Rockies, developer.nlr.gov). Inputs are the site's location, DC size, inverter (AC) size, tilt, azimuth, and mounting type (fixed or tracker). We use PVWatts' standard 14% system loss assumption unless the owner gives better values. PVWatts runs these inputs against a typical meteorological year (TMY) from the NSRDB to get **typical monthly AC output (kWh)**.

### *2. Weather adjustment: what it should have made in the real weather*

A typical year is not the year that actually happened. For each month we take satellite-based solar irradiance (global horizontal irradiance, GHI) for the site from **NASA POWER** (power.larc.nasa.gov), both for the actual month and as a long-term average for that calendar month. Then:

> **Expected kWh (month) = PVWatts typical kWh × (actual GHI ÷ long-term average GHI)**

Using one data source for both the actual and the average keeps the ratio consistent. A month that was 10% sunnier than normal raises the expected output by about 10%, and a cloudy month lowers it.

### *3. The gap*

> **Lost kWh = Expected kWh** – **Actual metered kWh**
>
> **Estimated lost $ = Lost kWh × the site's $/kWh** (PPA rate or retail rate. Where neither is known, we use the EIA state average commercial rate and label it as an estimate.)

Actual output comes from the owner's meter or monitoring export, or from public EIA-923 monthly generation.

**Normal aging.** Solar panels lose a little output each year. When we know the year a site went into service, we lower expected output by 0.5% a year, the median degradation rate found in NREL's review of field data ([Jordan & Kurtz, 2012](https://www.nlr.gov/docs/fy12osti/51664.pdf)). That way an older site isn't flagged just for its age.

### *4. What gets flagged*

Monthly models carry real uncertainty, so we don't flag a month for a small gap. A month is **flagged when actual output is more than 10% below expected**. We give the most weight to patterns, such as several flagged months in a row or a gap that keeps growing, over any single month.

### *5. Limits: read before acting*

• **These are estimates, not measurements.** Expected output is a model. Treat gaps under about 10% in a single month as noise.

• **Not modeled:** snow cover, utility curtailment, grid outages, planned downtime, inverter clipping beyond the PVWatts model, and site-specific shading. We check winter months and known curtailment before pointing to equipment.

• **Satellite irradiance** is an area average, not an on-site sensor. Local clouds, haze, or smoke can make it differ.

• **System details matter.** A wrong tilt, azimuth, or DC size shifts every month's expected value. We list the inputs we used so the owner can correct them.

• **Possible causes, not diagnoses.** Monthly data can show *that* a site is under-producing and *when* it started, but not *why*. Possible causes to check include inverter faults, string or combiner outages, soiling, new shading, module degradation, and data or metering errors. Confirming a cause needs inverter-level data or a site visit by the owner's O&M provider.

### *Data sources*

• NREL PVWatts v8 API: https://developer.nlr.gov/docs/solar/pvwatts/v8/

• NASA POWER (monthly and climatology GHI, ALLSKY_SFC_SW_DWN): https://power.larc.nasa.gov/

• EIA-923 monthly plant generation: https://www.eia.gov/electricity/data/eia923/

• EIA electricity rates by state: https://www.eia.gov/electricity/state/

PVWatts weather data: NSRDB PSM V3 GOES tmy-2020 3.2.0.